User Funnel Analysis - MetroCar Ride-Hailing App

1. Project Overview

MetroCar is a ride-hailing platform that allows users to download the app, register, request and complete a ride, make a payment, and leave a review.

This project presents a comprehensive **user funnel analysis** aimed at understanding customer behavior, identifying performance bottlenecks, and providing actionable recommendations to enhance conversion, retention, and user experience.

The analysis focused on four **key objectives**:

1. **Build the User Funnel:**
   Define each stage of the customer journey - *Download → Signup → Ride Request → Acceptance → Tide Completion → Payment → Review* - and calculate conversion rates between them.
2. **Identify Drop-off Points:**
   Detect the weakest funnel stages and examine possible influencing factors.
3. **Evaluate Platform Performance:**
   Compare engagement, conversion, and revenue across **Android**, **iOS**, and **Web** platforms to determine where marketing and product improvements should be prioritized.
4. **Analyze User Demographics:**
   Assess the activity and conversion levels of different **age groups** to identify which segments are most engaged and most likely to complete rides.

Overall, this analysis provides data-driven insights into **where and why users churn**, enabling strategic decisions to improve MetroCar's growth and customer satisfaction.

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2. Data Overview

The analysis was based on several datasets representing different stages of the **MetroCar** user journey.

Data was extracted with **SQL in DBeaver**, validated and visualized in **Excel and Power BI**.

Data Sources

- **app_downloads** - app installations and platform type (Android, iOS, Web)
- **signups** - user registrations with age group and session identifiers
- **ride_requests** - timestamps for request, acceptance, pickup, and cancellation
- **transactions** - completed rides and total purchase amount
- **reviews** - user ratings and feedback

Data Model

Tables were joined by common keys:

- `user_id` - links **signups**, **ride_requests**, and **reviews**, enabling user-level tracking.
- `ride_id` - connects **ride_requests**, **transactions**, and **reviews** for ride-level analysis.
- `session_id` / `app_download_key` - links **app_downloads** with **signups** to identify platform origin.

A Power BI data model enabled cross-filtering by **platform** and **age group** for deeper funnel analysis.

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3. Key Metrics

This section presents the main metrics which highlight a critical aspect of the user funnel and business performance within the MetroCar platform.

- **Total Revenue: $4.47M**
  Represents the total revenue generated from completed rides during the analyzed period. Indicates overall platform scale and financial performance.
- **Active Users: 6,233**
  The number of users who completed at least one ride. Reflects the active customer base and overall engagement.

- **Conversion Rate (Sign-up → Completed Ride): 35.37%**
  Indicates the proportion of registered users who successfully complete a ride. Reflects the efficiency of the user funnel.
- **Cancellation Rate: 41.98%**
  Shows the share of rides canceled before pickup. Highlights operational and user experience challenges.
- **Average Rating: 3.06 / 5**
  Represents customer satisfaction based on post-ride reviews. A relatively low score indicates inconsistent service quality and potential driver-related issues.
- **Average Ride Duration: 52.6 minutes**
  Indicates typical trip length. Suggests that many users rely on MetroCar for medium to long-distance rides.
- **Average Wait Times:**
  - **Request → Acceptance: 6.89 min**
    Measures how quickly drivers accept ride requests. Longer acceptance times can increase cancellation risk.
  - **Acceptance → Pickup: 8.50 min**
    Shows the time it takes for a driver to reach a passenger after accepting a ride. Directly affects user satisfaction and completion rate.
  - **Request → Cancel: 12.55 min**
    Represents how long users typically wait before canceling a ride request. Strongly correlates with user frustration and drop-off.

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4. Key Insights

1. User Funnel Analysis

The user funnel illustrates the full journey within the MetroCar platform - from app download to review submission. The analysis reveals that the most significant drop occurs between the "**ride accepted"** and "**ride completed"** stages, where almost **half of the users (−49.23%)** abandon the process after acceptance. This suggests potential issues with wait times, driver reliability, or lack of communication during pickup. Additional filtering by **age group** and **platform** confirms that this pattern remains consistent across all segments - meaning the issue is systemic rather than user-specific.

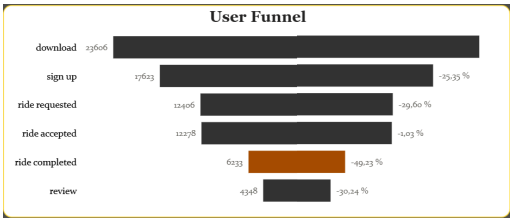

Figure 1. User Funnel Overview - *Conversion flow from download to review, highlighting a 49% drop-off between ride acceptance and completion.*

2. Identify Drop-off Points

This section explores **where and when users are most likely to abandon a ride**, and what operational factors contribute to the cancellations. The goal is to identify key drop-off points beyond the user funnel - particularly around ride acceptance, pickup, and waiting time.

Ride-level Conversion

The ride funnel reveals noticeable losses at multiple stages of the journey -from **requested** to **completed** rides. Out of all ride requests, **64% are accepted** by drivers, while **58% are ultimately completed** , indicating that a large portion of users abandon their rides even after acceptance.

At the same time, **not all requested rides are accepted** in the first place, reflecting limited driver availability during peak demand. In total, **161,825 rides were canceled** , which highlights a significant operational challenge that directly affects user satisfaction and platform efficiency.

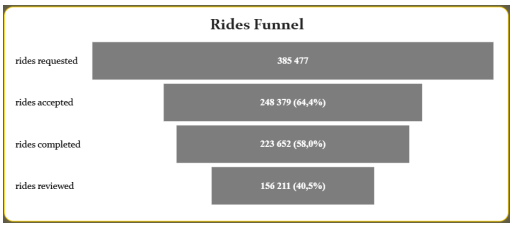

Figure 2. Rides Funnel Overview - *Conversion from requested to completed rides, showing that only 58% of requested rides are eventually completed.*

Cancellations by Time of Day

Cancellations are concentrated during **peak commuting hours (8-10 AM and 4-7 PM)** - the same periods with the highest number of ride requests.

This indicates that **service demand exceeds driver availability**, causing longer wait times and higher cancellation rates. The correlation between requests and cancellations suggests a capacity bottleneck during busy periods.

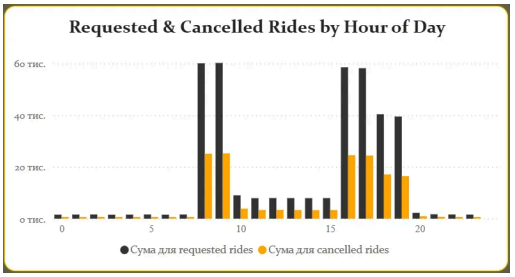

Figure 3. Requested and Cancelled Rides by Hour of Day - *Morning and evening peaks show the highest cancellation volumes.*

Impact of Wait Time on Cancellations

There is a clear positive relationship between **pickup wait time** and **ride cancellations**. When the driver takes longer than six minutes to reach the passenger, the number of cancellations **nearly doubles** compared to rides where pickup occurs within three minutes. This pattern highlights how operational delays - not user intent - are the main cause of drop-offs.

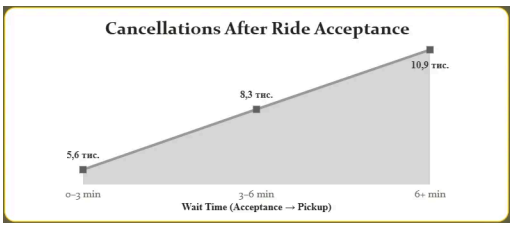

Figure 4. Cancellations After Ride Acceptance - *Longer pickup wait times directly increase cancellation frequency.*

Cancellations by Day of Week and Wait Period

Analysis by weekday shows a **consistent cancellation pattern throughout the week**, with almost all cancellations happening when users wait more than **10 minutes** for acceptance.

This consistency across days reinforces that long waiting times are a systemic issue, not a timing anomaly.

Cancelled Rides by Wait Period	Day of Week							
Wait Time (Request → Acceptance)	Mon	Tue	Wed	Thu	Fri	Sat	Sun	Totals
0-5 min	0	0	0	0	0	0	0	0
5-10 min	317	312	310	294	316	316	327	2192
10+ min	22746	22883	22769	22852	22771	22792	22820	159633
Totals	**23063**	**23195**	**23079**	**23146**	**23087**	**23108**	**23147**	**161825**

Figure 5. Cancelled Rides by Wait Period and Day of Week - *The majority of cancellations occur after a wait exceeding 10 minutes, consistent across all weekdays.*

Additional analysis of driver activity

The distribution of completed rides per driver shows that **most drivers (over 10,000)** completed only **6-10 rides**, while very few exceeded 15 rides.

This indicates **low driver engagement and availability**, which likely contributes to **longer wait times** and **higher cancellation rates**, especially during peak demand hours.

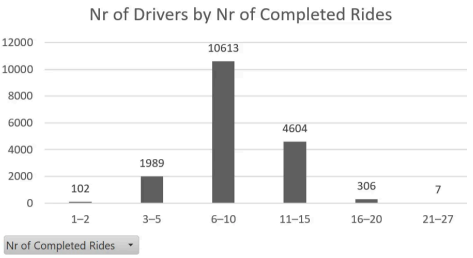

Figure 6. Driver Activity Distribution - *Most drivers completed fewer than 10 rides.*

3. Evaluate Platform Performance

This section compares **user behavior and revenue performance** across MetroCar's three platforms - **Android**, **iOS**, and **Web** - to understand where the product performs best and where optimization efforts should be focused.

Ride Activity by Platform

iOS users also account for the highest number of ride requests and completions. This indicates that **iOS users are both highly engaged and high-value customers.**

Android users contribute a smaller number of rides, while the **Web platform remains marginal**, with limited engagement - likely due to user preference for the convenience of mobile apps.

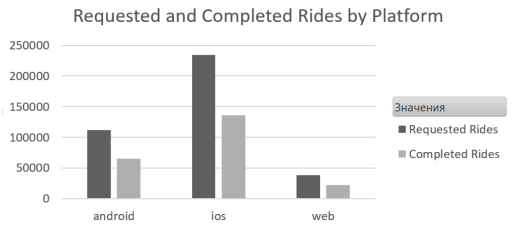

Figure 7. Requested and Completed Rides by Platform - *iOS leads in both ride requests and completions, suggesting stronger engagement among iOS users.*

Revenue Distribution by Platform

iOS users generate the majority of total revenue - **$2.72M (60.86%)**, followed by **Android users with $1.31M (29.24%)**, and **Web users with only $0.44M (9.9%)**. This indicates that **mobile usage dominates** MetroCar's ecosystem, with iOS having a significantly stronger commercial impact.

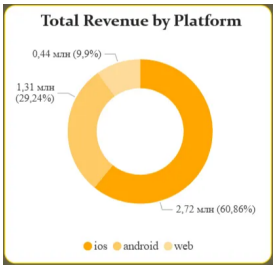

Figure 8. Total Revenue by Platform - *iOS generates 60.9% of total revenue*

4. Analyze User Demographics

This section examines how different **age groups** contribute to MetroCar's activity and revenue, highlighting which user segments are the most valuable and engaged.

Ride Requests and Completions by Age Group

The **35-44** and **Unknown** age groups show the highest engagement, accounting for the largest share of both ride requests and completed rides. The **25-34** segment also performs strongly, confirming its position as an active and loyal user base. In contrast, the **18-24** and **45-54** age groups show lower participation, suggesting less frequent usage and reduced dependence on the service.

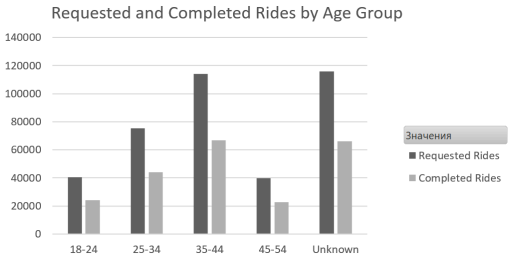

Figure 9. Requested and Completed Rides by Age Group - *The 35-44 and Unknown segments record the highest ride activity.*

Revenue by Age Group

The **35-44 age group** generates the largest share of total revenue at **29.89%**, followed closely by users with **unknown age (29.44%)**. This pattern indicates that **middle-aged users (35–44)** are the most valuable segment, likely due to higher spending power and more frequent rides. However, the large "unknown" category reveals a data gap - not all users provide demographic information, which limits targeting precision.

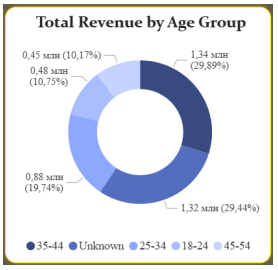

Figure 10. Total Revenue by Age Group - *Users aged 35-44 generate the highest share of total revenue.*

5. Analyze Customer Feedback

This section focuses on **user satisfaction and service perception**, based on customer reviews and rating distributions.

Most Frequent Review Keywords

Review analysis reveals that users most frequently describe rides as **"comfortable," "friendly,"** and **"reliable,"** indicating strong performance in driver behavior and overall safety perception. However, recurring negative terms such as **"rude," "uncomfortable," "late,"** and **"terrible"** point to **issues with punctuality and occasional poor driver conduct**.

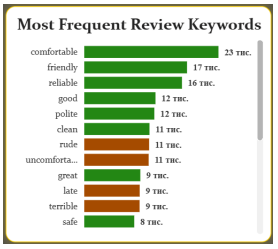

Figure 11. Most Frequent Review Keywords - *Positive mentions dominate, but negative feedback highlights key friction points.*

Customer Ratings

Customer ratings show a **mixed satisfaction level**, with most users providing mid-range scores.

Ratings of **4 and 5 stars together account for just over half of all reviews**, while **1-star ratings (nearly 30%)** represent a significant portion of feedback.

This distribution suggests **inconsistency in service quality**, where positive experiences coexist with notable dissatisfaction.

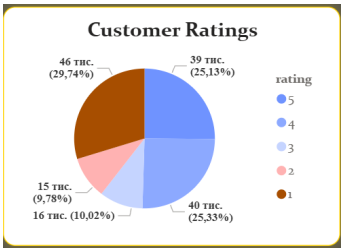

Figure 12. Customer Ratings - *The average rating is 3.06, indicating room for improvement in user satisfaction.*

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5. Recommendations

Based on the analysis of the user funnel, ride performance, platform activity, and demographic trends, several key actions are recommended to enhance product efficiency, marketing strategy, and overall customer experience.

1. **Optimize Ride Completion and Reduce Drop-offs:**

   Implement real-time driver allocation and dynamic dispatching to minimize waiting times between ride acceptance and pickup. Reducing pickup delays can directly lower cancellation rates and increase user trust in the service.

2. **Improve Driver Engagement and Availability:**

   Introduce performance-based incentives and flexible scheduling for drivers to encourage higher participation, especially during peak hours. Better driver retention and engagement will help balance supply with demand and reduce service delays.

3. **Enhance In-app Communication and Transparency:**

   Provide users with accurate, real-time updates on driver arrival times and delay notifications. Clear communication can reduce user frustration and perceived waiting time, improving the likelihood of ride completion.

4. **Strengthen Customer Retention Programs:**

   Launch loyalty rewards and personalized offers targeting active users, as well as reactivation campaigns for those who recently canceled or became inactive. Tailored retention strategies can boost repeat usage and lifetime value.

5. **Focus Marketing Efforts on High-Value Segments:**

   Prioritize campaigns for **iOS users** and the **35-44 age group**, who show the strongest engagement and revenue potential. Use behavioral segmentation to create targeted promotions and app updates that resonate with these core audiences.

6. **Re-engage Underrepresented Age Groups:**

   Develop awareness campaigns and user-friendly onboarding for younger (18-24) and older (45-54) users. Educational content or simplified app features can help older groups adopt the service more confidently. Launch student discounts or flexible payment options to stimulate usage among younger, price-sensitive users.

7. **Improve Data Collection and User Profiling:**

   Encourage users to provide demographic information during registration to reduce the “Unknown” segment in analytics. Better user data will enable more accurate targeting, personalized experiences, and improved strategic planning.

8. **Enhance Service Quality Monitoring:**

   Track customer ratings and review keywords continuously to identify service pain points. Use this feedback to train drivers, optimize customer support, and proactively address recurring issues (e.g., “late,” “rude,” “uncomfortable”). Prompt users to share specific feedback categories (e.g., driver behavior, cleanliness, waiting time) for more actionable insights.

Summary:

By addressing these areas - from operational efficiency to targeted marketing and user experience - MetroCar can reduce friction within the funnel, increase retention, and strengthen its position in the competitive ride-hailing market.

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7. Technologies Used

This project combined SQL-based data extraction, spreadsheet validation, and interactive dashboard development to provide end-to-end analysis and visualization.

Tools and Environments

- **DBeaver (PostgreSQL):** used to write and execute SQL queries for data extraction, cleaning, and aggregation across multiple tables (users, rides, transactions, reviews).
- **Excel:** applied for intermediate analysis, data validation, and creation of pivot tables for early insights (e.g., cancellations by day and wait time).
- **Power BI:** developed the final interactive dashboard integrating user funnel metrics, ride performance, platform analysis, and customer satisfaction visuals.

Key Technical Skills

- **SQL:** complex joins, CTEs, aggregation, date-time operations, and conditional filtering.
- **Data Modeling:** building relational models in Power BI (one-to-many and many-to-many relationships) for dynamic filtering by platform and age group.
- **DAX Measures:** calculating KPIs such as conversion rates, average waiting time, and cancellation rates.
- **Data Visualization:** designing a clear, business-oriented dashboard with customized color theme, layout, and dynamic slicers.
- **Analytical Reasoning:** translating data patterns into actionable insights for product, marketing, and service optimization.

Example Queries and Formulas

- SQL Example - User Funnel Drop-off Calculation
  Calculates the number of users at each funnel stage and identifies where the biggest user drop-offs occur by platform. This query served as the foundation for the User Funnel visualization in Power BI.

```
WITH funnel_counts AS (
    SELECT
        platform,
        funnel_step,
      funnel_name,
      COUNT(1) AS nr_of_users
    FROM funnel_analysis
    GROUP BY platform, funnel_step, funnel_name
)
SELECT
    f1.platform,
    f1.funnel_step,
  f1.funnel_name,
  f1.nr_of_users,
  (f1.nr_of_users - lag(f1.nr_of_users) OVER (PARTITION BY f1.platform ORDER BY f1.funnel_step )) AS drop_off
FROM funnel_counts f1
ORDER BY  f1.platform, f1.funnel_step;
```

- DAX Example - Active Users Calculation
  Computes the total number of active users based on completed rides, used as a key performance card in the dashboard.

```
Active Users =
CALCULATE(
    SUM('Users Funnel'[nr_of_users]),
    FILTER(
        'Users Funnel',
        'Users Funnel'[funnel_name] = "ride completed"
    )
)
```

8. Conclusion

The analysis of MetroCar's user funnel, platform performance, and customer experience revealed that the greatest opportunity for growth lies in improving operational efficiency and user retention.

Most users drop off after ride acceptance, primarily due to long waiting times and inconsistent driver engagement. iOS remains the most profitable and active platform, while middle-aged users (35-44) represent the most loyal customer base.

By optimizing driver allocation, enhancing in-app communication, and refining user targeting, MetroCar can significantly reduce cancellations, strengthen brand trust, and improve the overall customer journey from download to review.

9. Dashboard Preview

This section presents selected screenshots from the **Power BI dashboard** created for this project. The dashboard integrates SQL-based metrics and DAX measures to provide a clear, interactive view of user behavior, platform performance, and business outcomes.

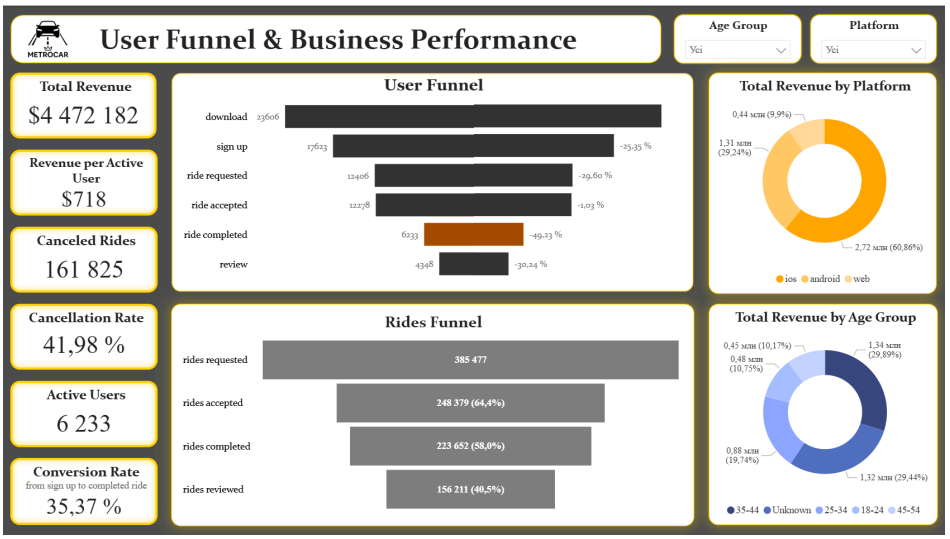

Figure 13. Power BI Dashboard for Metrocar - *Page 1. User Funnel and Business Performance*

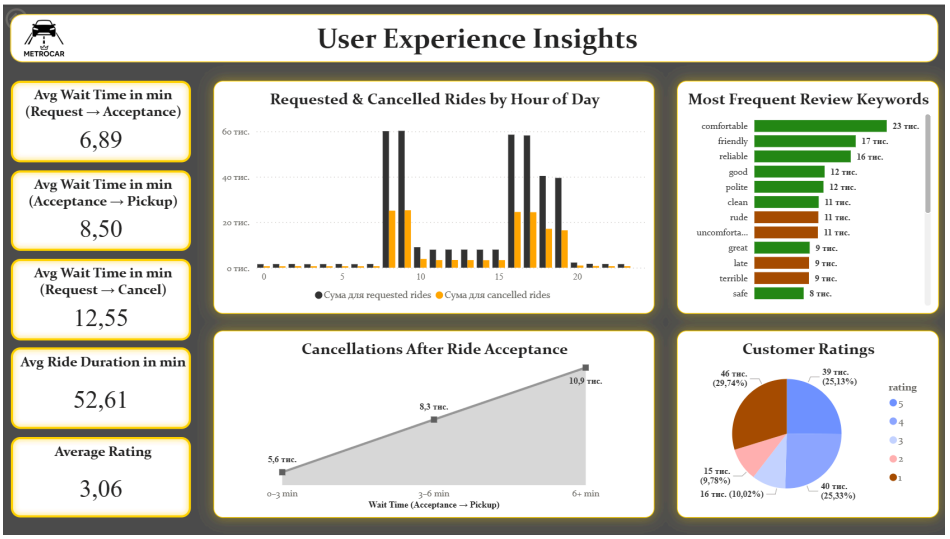

Figure 14. Power BI Dashboard for Metrocar - *Page 2. User Experience Insights*